4 – October Focus Group – FG2

Interviewer

And sort of similar to the previous ones, we'll have a bit of an introduction session just because people are swapping between focus groups, so you won't necessarily have met everyone in here before. This time. So as with before, just say the type one diabetes technology you're using, which wearable you choose or the smartwatch for this one, and then this month, we're gonna do sort of a little hack or gadget or little tip you have that you found has helped with your type one diabetes management. It doesn't have to be anything too in-depth it's just a little something that springs to mind. So, I'll go first. I'm using the t:slim and Dexcom G6 sensors. I had my t:slim fail at the weekend, which is a bit scary. The um. It just said operation failure and then refused to do anything, but it's kind of working again so that's something. And then I use a a Fitbit sense, which some of you might be using as part of this study, and I've been using it for a few years now as part of the project. And then my little hack called a trick that I found not not so long ago, which I'm sure people find out a lot earlier and kind of figure out but it took me a while. It's just kind of that use of exercise and that if I have high blood glucose levels then exercise can really like speed up the effect insulin has rather than necessarily just bringing it down on its own. Kind of using those in tandem. But yeah, then I'll pass on to [P10], if you happen to go next.

P10

Yeah. So I use again same as [Interviewer], so Dexcom and t:slim. I'm on the Versa 4 Fitbit, which is quite similar to the Sense I believe. And kind of the kind of hack I did, I managed, I assume other people have found out as well, but you can get your Dexcom readings on the Versa and the sense too, I think. You just need the right watch face. But I found that quite useful, just having the wrist.

Interviewer

Yeah, a very relevant point. Do you mind um, picking out someone to go next and introduce themselves?

P10

Um, [P18].

P18

Hi, I'm [P18]. I am on the Medtronic system and I have now got the sensors that go with them, the Guardian 4. I'm using an Apple Watch and I don't really have a hack actually. I wish I did, probably something similar to what you've already discussed with the exercise [Interviewer] and then obviously about linking them to your smart devices with the sensors. Did you want me to pick someone else?

Interviewer

Yeah, that'd be great. Thanks.

P18

Cool, [P01].

P01

Hi, I'm [P01]. I'm on the Metronic 360 I think and the FreeStyle Libre 2 and I've got the Fitbit Luxe. In terms of tips, yeah, I'm not really sure for one, like with the technology, but I guess it kind of helps with the technology. I found that you can get this stuff called Second Skin, which is really good for sticking on the sensor when I do like swimming or other exercise that might knock it off. And I found that better than like tape and stuff. Um, [P22].

P22

I'm on the t:slim and Dexcom with the Versa 4. I don't know if it's really a hack again, but I do the same walk at the same times to and from uni everyday. And using the Fitbit I found it useful with tracking my steps at at the, with the routine, and then also being able to control my levels a bit better at that time. [P24].

P24

Yeah. I'm sorry I can't put my camera on. I don't have good Wi-Fi at all. I use the t:slim with Dexcom G6 and I have the Fitbit Luxe. And you're all frozen, so you probably can't hear me. I definitely can't see any of you moving.

Interviewer

We can still hear you [P24], at least at the moment.

P24

Ohh hello. I heard you again.

Interviewer

Hi, are we coming through is that? OK, yeah, that might work be easier. And while [P24]'s doing that, I'll try and load up this whiteboard feature. Um. In fact, see if I can load up the previous one and just save editing. Yeah. So has that kind of the whiteboard appeared for most people in terms of what's there. You should all be able to edit it. Umm, it's just to be there to use the little prompt if you think it will help with you and try to kind of just explain things. Um. Brilliant. And there's {P24]'s

introduction. Ohh nice yeah the using Siri to be able to read out your blood glucose levels. Yeah, I think that does sound very cool. Very futuristic. Um. No, great. So yeah, that's kind of everyone introduced. There might be one person who joins meeting a bit later, we'll bring them in as it goes. Yeah. As I said, this whiteboard is here just to help us as we're going because the first point I want to kind of kick off discussion with is around this idea of ideal or ideal. Well, the idea is around BG profile and what I mean by BG Profile is kind of this view that often you'll get from a CGM sensor. So FreeStyle Libre when you scan it or off a Dexcom where you have that sort of background line that kind of tracks where your blood sugars have gone throughout the day. Um. And so from this I really want to think about what the people's like ideal, what people would say that the ideal blood glucose level is? And if that's the same as the blood glucose they aim for or whether there are variations between the two. So, if anyone's happy, just maybe start things off with that.

P10

Uhm, I can start if that's alright. Uh. So, the kind of levels I aim for, well, it's changed. So, when I was on pens umn, I found I was doing quite an odd thing where I was really kind of having loads of Levemir so I could run quite low during the day and it worked quite well. So, I was normally aiming for like 4/4.5 was where I felt was quite comfortable. But, now on pump I aim for about 5/6 ish, but then if I'm out and about. So, like if I'm at uni or like in lectures or bits like that I do tend to aim for like 7. So, it does vary quite a lot, but if it gets much above that, I tend to start correcting. So those are the kind of the levels I am for. But ideally, I feel more comfortable at 5 at the moment I think.

P24

I'm similar. Um. I aim for about 6, because 5.5 feels too low. It feels like something could happen, and I'd immediately be low, whereas if it's 6, I've got a bit more time to to react to that. But like if I'm about to go to sleep, 5.5 is great.

P22

I think I'm the same as [P10]. The whole time, just between 5 and 6. And then when I'm out and about, it's more 7 and I find with the t:slim that I can actually, well obviously not all the time. But majority of the time I am actually able to stick to that. So yeah.

P18

When I'm like, just doing day-to-day bits and pieces, I like to run around 6, but then if I'm exercising. If I'm exercising, then I've got to run about 10 just because otherwise I'll crash even with the sensor and stuff, but then if I'm not working, I like to run around my 7/8 really because it's quite an active job.

P01

Yeah, I think I I like to run around sort of 5 or 6. In reality, I think mine like varies load, but if I'm like exercising or if I'm about to go to bed, I tend to keep it a bit higher. Or also if I'm like about to do something where I I know I won't be able to check my blood sugar for ages then I tend to run a bit higher just because I feel like it's it's safer short term, although it's probably really bad long term.

P24

I do that as well before a long drive, I make sure I'm running a bit higher because if it goes low I have to stop driving. If it goes higher, I just feel a bit shit.

P01

Yeah, that's a great example actually, cause it's like you have to choose between what is going to potentially get you, like legally in trouble because you can't obviously drive when you're hypo. Whereas you can, you could drive if you were like HI, legally, but you feel terrible, so it's it's hard to get the balance, but I tend to like aim for a bit higher from driving as well.

Interviewer

Yeah, it seems like. You're you're all commenting on that, that kind of level you want to be at varies throughout the day or based on the activities you're doing. I think it, yeah, a couple of people referring to quite an interesting point that sometimes that you kind of put it higher just because of the activity you're doing, even though you know there's kind of complications associated with having readings at this level. I wonder if people are happy to kind of talk a little bit more about how they feel about doing that and how they weigh that up in themselves.

P22

With the driving example and with exercise, the thing is I think like what was previously said, I'd much rather be a bit higher than low. Especially driving like on the motorway or something, or doing like getting in the flow of exercise and then like having to stop. It just obviously like disrupts everything, so I personally would much rather have a bit of a headache for a little bit and be able to carry on what I'm doing rather than shooting down low and then having to spend however long by the side of the road or stopping what I'm doing. Yeah. So that's how I weigh it up.

P24

For sure, it's those moments that diabetes is most like. Like, that's when I get really emotional, when it interrupts something I want to do, and it's just because I'm diabetic that I have to stop doing something. That's when I get really angry and upset. Um. I'm not gonna lie, I don't really think about the long term complications, especially with driving. Like at the end of the day, if my bloods go low, there's a high, much higher chance that I'll end up crashing or doing like, as in crashing my car, not my bloods. And like, that's so much more dangerous than potentially having like some long term health implications of are high for a few hours.

P01

Yeah, I I relate a lot to sort of feeling angry about that. Um. I think for me as well, it's more like I I feel more angry when I think about like the long term complications as well because I hate that like I'm having to put myself in a situation where I'm potentially like worsening my long term health just because like I don't have time to like, stop and do it more carefully or whatever. Particularly if it's like a a work thing, I feel like then I have like resentment towards my job if it's like forcing me to make sort of decisions like that, that if I wasn't having to do that, I I might be able to like prioritise my health a little bit more. But yeah, it's complete.

P10

I'm, it's somewhat similar, but I always find myself like even at school or like uni with like lectures and things like that. It's not particularly self-conscious about it, but I hate beeping, so I always run slightly higher and that's the kind of thing that annoys me most from time to time because like exercise is fine because, I agree with everyone, it's kind of like you do get the benefit of it's much less dangerous when you run high. But if I'm running slightly higher just say like I don't need to beep or fiddle around the middle of a lecture but like it's not dangerous or very kind of low because I have jellybabies on me and like it doesn't cause that much disruption as well, as long as I'm not really like, so I like use concentration. But I think that's the kind of time that I'd get a bit more frustrated because it's a choice I'm making and it's a choice I'm aware I don't need to make. But, it's just kind of like for ease of right not self-conscious, but just as I find it annoying if I suddenly start beeping and everyone else is silent. It's kind of a bit distracting.

P01

Yeah, I think that's a really good point. Like it's you're essentially doing it for the benefit of other people, so that they don't have to have the disruption of it, even though it's probably more disruptive for you for feeling bad. And I definitely feel like self-conscious of the alarms as well because it it just attracts so much attention.

P22

I completely get that because I I feel overall very confident about my diabetes and being very open about it. But I remember my very first day of uni a couple years ago. So, I do drama, so it's everyone like up on their feet in a room together. And it was the very first day, the very first session ever, in the first like 2 minutes we were all up on our feet and my pump started beeping because I was low and everyone turned around to look at me and I just remember thinking like, “ohh like why did I put so much insulin in for the, like, the my breakfast or whatever?” And then I that my first thought was “ohh, like it's so embarrassing, whatever”. But it shouldn't be embarrassing. But yeah, no, I completely completely get what you're saying about that.

P18

Yeah, I was just going to say the same. It's like they come from a good place because you you do have the alerts and then they're always like, “are you OK?” Like all your colleagues saying, like, like, “what's up? Is something we can do?” But it is that sort of drawing attention to it when in actual fact you could do with being a little bit higher and sort of being a detriment to your long-term health, a little bit more. Just to sort of avoid that embarrassment, or like having to talk about it or reassuring people that you're OK, you're not going to collapse on them whatever they like to describe it as so yeah.

P01

Yeah, people are always so worried about you like collapsing on them. And it's like, “I don't have control necessarily over that”, like over how bad it is. Yeah, it's. It does definitely make it sort of awkward.

P24

Yeah, I have the same experience of like that, anxiety around it beeping. I've turned off all my alarms on my pump, apart from the one that you can't turn off. I just have them buzz instead of beep. Um. But but even then, I'm always so anxious, like, “oh, my God, is it going to tell me that I'm urgent, low”, and then it's just going to beep and I can't control it. And it's so loud and it happened to be in a cinema recently and it was really embarrassing. And it's like that element of, like, nobody actually cares but you're thinking to yourself like, uh, I bet someone is thinking “oh, that stupid idiot should have turned their phone off” or something like that. And I don't know why I care about that because I know in myself and my friends know that it's just my pump. But it's just that, like social anxiety element of it. I'm quite lucky that my friends don't do that whole thing where it's like, “oh my God, are we gonna? Are you gonna die on us are. You gonna collapse on us?” Like I'll just be like, “oh, I'm dying” and whenever my pump beeps and then they know that it's chill because I'm joking about it. But like all the other stuff really like, I definitely relate to.

P01

I I think the thing that adds to it for me is about like the social awkwardness is the fact that like being diabetic, especially when you're like a child, it's always like an extra like thing. Like when you go on school trips and stuff, it's always like you're you're the annoying like kid that has all the has to have the extra fuss around and if you're like like. Obviously everybody's different but if you are like a bit awkward and socially awkward anyway, then that just makes it so much worse. And like, you just want to be normal for once. And yeah, the alarms definitely they're really, really helpful, make you feel not normal.

Interviewer

Yeah, I mean, just to maybe clarifying that point a little bit, is that in terms of what blood glucose level you're aiming for then? I mean, sounds like people aiming for a bit higher, is that to avoid they feel they have these implications more from like hypos rather than hypos or is it different for different situations or something else?

P01

I think for me, I really sorry you.

P10

I was. No you go.

P01

Yeah. Oh, OK. I I was just gonna say I really. I hate the feeling of being high probably more than the feeling of being hypo because I. I I don't know it it depends on the situation but I hate feeling high. But I feel like it's less of an embarrassment and less of a pain to other people because I sort of feel a bit under the weather, but I can like, put a brave face on and cope with it and seem normal. Whereas, depending on what I'm doing, obviously there's the situations like driving where being hypo is actually dangerous, but also like if I'm doing a presentation and I need to look professional, if I fumbling over my words and acting hypo then it looks worse to other people than just being seeming a bit tired, maybe.

Interviewer

Yeah. [P10], did you have a comment on that as well?

P10

Ohh, you know it was just basically that I was. But yeah, I would run higher basically because I can't really. I never really feel high in the same sense like I can't really feel it. I can taste it, kind of my mouth feels weird and if I get really high, I do kind of start to feel it. But I always run a bit higher because then you don't beep at all because obviously I'm quite similar. So, all my alarms are turned off to just vibrate. Obviously, apart from the urgent low, which you can't. But like the high ones, I never, they don't beat. So, it's kind of like when they do have the kind of bits like going to the cinema or examples like that where you like everyone's quiet every turn their phones off, I always would run higher because even if I do get really high, it wouldn't beep in the same way. It would just vibrate so that's just I can still deal with it. And I still like deal with it when it does alert me, it's just it only alerts me, I think it's the kind of logic partly behind the higher rather than lower.

P22

I feel the exact same about the the alarms. I have all my alarms on vibrate as well and I just wanted to say that I didn't know that there were other diabetics who did the same thing. So, it's quite comforting to hear that there are people that hate the alarms just as much as I do.

Interviewer

Yeah. And it's interesting to hear that kind of like range of different reasons for what adjust the level you're aiming for at different times so so there is clear that social implication of different effects also interrupting behaviour and limiting you, preventing you from doing things like driving. What about the effect of doing something that you have done often or done many times before versus trying something new. How does that change what blood glucose you aim for?

P24

I kind of free ball it on the first time. Like I just see how it affects me if I'm doing something new, unless I know exactly what it's going to be like. Like say if I was like randomly about to go and run a marathon, that's quite an extreme example. Say if I was about to go and run a mile, I wouldn't. I've never done that because I don't run, but I'd be able to know to run a bit higher. But if it's something wild like, I don't know, jet skiing, I have no idea how that's going to affect my blood, so I'm just going to see how it goes.

P22

Yeah, I think with stuff that I sort of do every day, I can aim to be a bit lower like between 5 and 6 where I'd like to be. But, with something completely new, I would definitely try and be, when I say high, like not super high, like probably around 9/10, just to be on the safe side. But then once I'm used like if I do that activity a lot, then I can get comfortable and know the routine and be able to go back to a lower level. Um, just takes time I think.

P18

I'm the same as well, but I don't follow like the diabetic rules when it comes to exercise, particularly, because I do rely on the sensors telling me where I'm at. But for like if I was starting a new sport or an exercise that requires that pump to come off for like and water reasons or pressure reasons um then I would be very inclined to like follow those exercise rules. Like stopping every, I think it is every half an hour, and testing and having a carbohydrate snack. But yeah, it's really variable I'd say. Like the ones I'm, the stuff I'm used to, I'm just a little bit blasé with it. I just keep going.

P10

Yeah, it's, I always find it's quite, not a dangerous combination, but I always, say anything I run higher, I always try to. Um. Especially because I just started uni like 2 weeks ago, so like I've done quite a lot of kind of societies and bits like that. But all at the first session for every single one of them I've ran like at 9 or 10. And sometimes, like if you're nervous for it, I then like the adrenaline, I get quite affected by that and it does keep me higher anyway. So, it is kind of, I know it's coming and I still kind of do it anyway, so it's then slightly harder to get to during activity. I'll give myself insulin and I just won't go down at all because I've got adrenaline in my system and then after I will go down normally, but it's just I'm high for the entire time. So, that's at like 10 or 11, which for me is actually. I don't normally go to these kind of things from day-to-day, but I I still choose it just because it's easier that than if you stop something new. I think especially it's a game kind of both the not knowing how

low I might go, how active an exercise is or like how active an activity is, but also the kind of again social aspect. I'm meeting new people, the first thing I want to happen is I don't want to just have a hypo for the first time meeting people. So, that's another, is again the social side, but it's a part of the reason why.

P24

I guess there's also like that that, when you when you say doing something new, I guess you could also read that as like doing something you do regularly, but your routines been changed a bit. So, say if I've bought a new cereal or had toast for breakfast instead of cereal, so it doesn't give me that spike in sugar. I can eat my breakfast, think great, “I have carbed for that perfectly, I know exactly how many carbs are in that, I've got that right”. And then I can go start walking to uni and my bloods are plummeting. And I it's cause I've forgotten the cereal affects my bloods in different ways to toast. Or like I've had breakfast at, I've been hungover and had breakfast at 1:00 PM instead of like 9:00 AM, so my carb ratios are different and I didn't realise they were wrong at that time of day. It's like there is little changes of routine when you don't even realise that it's different until it hits you. Sometimes you just cannot, you cannot prepare for it.

P01

Yeah, I also think like even things like that that are out of control, like I tend to have a pretty standard routine like week to week and I go to like the same exercise classes each week. And I know like for each different one how much carbs roughly I'll eat before to like keep myself stable. But then randomly one week the teacher will like decide to do a much less intense class because she's injured, or like she'll do a more intense class because, like, there's less people and she can focus more on us or something like. And then it just completely screws up like my my plans and what I think has been like being really well prepared for it.

Interviewer

Yeah. So, I think from all of that, there's lots of these occasions where these changes are usually an increase in what the blood glucose levels your aiming for. I was wondering how easy you find doing that with the bits of technology that you currently use?

P22

I've had a couple of pumps and I have to say that the t:slim and with like the control IQ with the Dexcom is the best combination that I've ever used. I had. Er. I've had Omnipod and Medtronic and both of them, although they were good the time of my life that it was, for now, for being at uni and for have a fairly unpredictable schedule, they save my life basically. Especially with things like like drinking, which I don't know if I was on injections, I don't actually know how I would do that and go to sleep feeling safe. But with the, Dexcom and t:slim like in that sort of situation and other similar situations, I feel like 90% of the time I, well I feel comfortable putting my life in the hands of my technology.

P24

I'm the same. I like I've been on a pump for uh all of my diagnosis, apart from one year when I got diagnosed. But even just the comparison between my Medtronic and my second Medtronic and the t:slim, like I can trust the t:slim I I can let it do its thing. I've had so many nights out where like I wouldn't have remembered to correct for a high or to have some sugar for a low and it's become less severe because of the t:slim, like it's it's just brilliant.

P10

Yeah, no, I'm quite similar, so I. It takes a lot of the pressure like you not having to worry about it. So, I'm on t:slim as well. So, like again when I'm drinking bits like that, like I don't mind at all. And before I used to get quite, not often, but I used to get some quite bad like nightly lows and I don't wake up to my alarms at all. So, like even for this summer, like I was on holiday and I hadn't bought enough kind of pump cannula replacements because I didn't. I'd forgotten that because I was going swimming they were going to fall off because, like, they didn't like it and I hadn't brought any kind of like extra sticky bits. So, I had to go on pen for a few days and I had one, really bad night time low. Like my mum had to call one of my friends because, like, I just wasn't getting up because I didn't wake up to things. But like being at uni and like not having that kind of back up, having the t:slim helps a lot because I know it just like I won't get night time lows in any of the same way. But kind of on a kind of counterpoint, I do find that when I do try and run high for things like lectures, it is less flexible. Because obviously the control IQ is great but sometimes I do find that it kind of overreacts too much, and I know that's just the ratios. It's just because I've been ill like freshers flu and bits like that. So, like I've had to change quite a lot of my ratios quite quickly. Like it seems every three days, I need to change most of my settings. So, I know it's mostly that, but sometimes it does mean like it's correcting too much and it's making me go low and sometimes I just kind of like stop it, you know, for a bit, so I know I don't go low, um to let myself get up a bit. Because I know the pump will probably correct too much and so there is the kind of two sides of things. It's small which helps, but sometimes you do want to override it, which it doesn't let you do.

P24

I have the same experience with nighttime lows um. Like no matter how much you do, sometimes it just, it just happens and like I don't wake up to my alarms really either because obviously we've all been hearing them for so long that they're just nothing now. And you can't change them like you kind your alarm to wake up, it's just like you get used to the beeps. Um. And loads of times I've woken up because my mum's been calling me because she has my bloods as well. My mum's been calling me or my flatmates like coming like "[P24], why is your phone going off? Shut up". Because they don't know what it is and it's like "oh, I could have died, great". And that's happened so much less than um, so much so, much less since I've had the t:slim. And like I had those adjust adjustment issues as well, especially when I started the t:slim, of like having to change my ratios so constantly. But, I would say for the past few months, I haven't had to change a thing which has been really nice, I think I've finally got into the just like the right to sort of numbers.

P18

I've, um like only recently gone on the technology that creates this closed-loop with the pump I'm on currently. Um. So, I'm still sort of getting the the ratios right, but I think it's really important for me to run myself a bit higher towards bedtime because of those lows. But in actual fact, what's happening now is that I am having a hypo right before bed and then like the natural rhythm of then a reoccurring hypo like later in, sort of like early morning because I'm so tired or I'm feeling the effects of the hypo so much like. I sort of switch off to the beeping or the vibrating because I'm so like tired or I'm feeling like so hypo. So, it's quite a vicious circle really, because I don't know about your guy's pumps, but mine vibrates for like a good 20 minutes and then eventually it's like this really horrible high pitch alarm, which eventually gets you sorted, but it you know it's a long time to be in a danger zone really.

P01

I'm still on like the sort of older technology that doesn't allow you to have like the closed-loop, and I definitely like night time hypos and also hypos like right before I go to bed are a massive problem for me. And I'm hoping my my pump renewal is in like February, I think. So, I need to choose between like the three different closed-loop systems that they fund at the moment, and I was really like set for going with t:slim because my brother uses it and I know so many people that use it and have a great experience. But the nurse at the hospital is like sort of really pushing for the Medtronic one and also this other one which I've never heard of which only works through an app which you can only get on an Android phone. So, I need to look more into that. It's it's like, really scary having to make such a big commitment to something that you're gonna be using for like the next 10 years, cause, see, once you get it, you can't really change it. But I think either, which ever one I go with, I'm so looking forward to getting a closed-loop because I think it will make such a big difference to my like daily control and and make just doing anything easier, I'm hoping.

Interviewer

Yeah. I think one thing within that I think the [P10] you were saying was around the, yeah, there being times where you want to kind of change it more or have more control and the pumps over correcting for you. I'd be interested to hear maybe more people's opinions on that, particularly with regards to the closed-loop and if they find they have enough control over yeah, adjusting the blood glucose level that it's aiming for, or if they're things they'd change about if they could.

P22

Uhm, I personally I don't. I haven't found an issue with that I think. Maybe from what I've heard from the conversation, I do feel more comfortable running at a slightly lower level so. Unless, like we were saying earlier that I'm doing exercise or driving, which with my uni schedule, neither of those things have a very much. But if unless I'm doing so one of those things where I do want to run a little bit higher, I don't really worry about that and then I find that with my settings and stuff that I don't know, maybe I'm just lucky, but it seems to work 90% of the time really well for me. So, yeah.

P01

One thing with oh, sorry. I was just going to say, obviously I've not been on it, on the closed-loop yet. But, you were asking about like the what you wish you could control more and when I'm like, looking into the different options, it seems to be that like different closed-loop systems have different restrictions on what you can set your target blood sugar as. And like some of them you you can't set it like particularly low, it has to be sort of between 3.9 and 7. And you can't make it lower, which is just sort of fine for the moment and day-to-day life. But one of the reasons why I'm concerned about it in the future is for like pregnancy. You're meant to have a lot tighter control, and if I'm choosing a pump for the next 10 years, then that's something I have to consider and I find it really frustrating that they don't like it, the pump manufacturers don't trust you enough to like set your own blood sugar target like. Surely that's an individual thing that everybody should be able to set to whatever they like, but it seems to be more restrictive than I would have hoped.

P18

Yeah. Sorry. Just to like follow on from what you're saying there. With my pump, I would like to be able to not only change the like increment, but whether you could do that over a 24 hour period and have different ones for different points of the day or like profiles that you could apply if you are doing exercise. So, like the Medtronic one, you can have a temporary target which is where you like you're around 8, 8 mmol/L, but that is more of like a suspending things, it stops giving insulin altogether. Then the minute you resume you can you can go low very quickly or in actual fact, you can go too high, too quickly. Um. But also, with this Medtronic one as well, is you can only set it as 6.1, 6.7 or 8 and you can't do anything else so it's it's crazy really. Because sometimes you want to be you want to be 10 or you want to be 9 but.

P10

You know, I'm quite similar. Sort of with t:slim, I think the lowest you can do is like 6.1 or something like that and it's not particularly low. Like so before I was like before I was in the pump, I was almost always like overnight, I used to run at 3.9 to 4 and just I used to stay the whole night. It is rather is I used to have a few night time hypos but like I just found I slept so much better when I was that much lower. Even going up to 6 like I do notice that my sleep is slightly worse. Just when I'm that much lower, I just have a deeper sleep. So, it is one of the things I really kind of miss about having full control because with the control IQ like I will just be at 6. And then because it's kind of slightly higher and we find that it doesn't, it's not as much of a flat line. But yeah so. But I I do believe I have heard well. Well because I got my pump quite recently, just in January, I did hear that I think it will eventually roll out that you can. Yeah, I think it's happening in America at the moment. I don't know you know that for certain, but I'm quite looking forward to that update, if it does come.

Interviewer

I I think that's a a range of interesting views on it, and I particularly interested in that yeah, kind of that trust element from companies and that it feels like they won't allow you to do what you want to do or don't trust you to have that control. Um. Maybe as part of that, and I think the were a couple of reference to it, is like how accurate you would say the information you put into the system. So, whether that's maybe a carbohydrate value and if you'll adjust it yourself because you don't want the system to be aware of how much carbohydrate you've actually had or whether you want it to think

you've had more to give a different dose. Yeah. I just wonder if anybody has like particular occasions where they've done similar to that around that accuracy of the information.

P10

I find.

P18

I probably like adjust my bonus every time by couple of increments and it like it's almost like innate in me to literally change it. Um. Because I do feel like it is off, but it it probably isn't, it's just like that fear inside you to to alter, to not alter it because it's too mechanical or it doesn't know you enough.

P10

Yeah, no I'm quite similar. It's always kind of not, it's a careless word to use, but I do it based on a vibe, which in my life, just like I know better than it does. Like the closed-loop system's really smart, but like, I know, if I didn't walk 15 minutes ago, I need ever so slightly less. So, I always find myself like 9 times out of 10 I will slightly override, based on kind of the exercise I've done. Because obviously it doesn't include that in the same way. So, I do find myself overriding the pump quite a lot and and I do like that you could do that quite easily. You just kind of click it and you can change the units. So, I always start off with doing the right amount of carbs and with that lot and my ratios are fine. So, like if I haven't been doing anything all day I know that I do just go straight with what it suggests, and I always look at what it suggests first. And then I kind of think to myself, "OK, do I need more or less?" But I do change it quite often.

P01

Yeah, I I think there's always going to be times where, like you do know better than the pump, because no matter how smart it is, it it can't know like what your plans are. And so like, if I'm eating lunch and I know I'm about to go on a massive hike, then I'm going to put less carbs in or I'm going to reduce the bolus. And there's just like, no possible way that the pump would ever know that, because I've not done it yet. It's just future plans.

P24

I definitely um adjust mine sometimes. Not like. I would say that the majority of times I do just leave it. But oftentimes it's when I'm high if I want to be high, I won't do the added correction. Or if I've just gone ham with like, every 5 minutes I've been correcting, correcting, correcting and then it's dinner time, then I won't add a correction onto my dinner dose because it's just gonna get confused. And I have a really bad habit of just like microdosing the corrections, it'll be like 0.1 units and I'm like, "yeah, I need that". So, I definitely think like there are some times where, it's usually when my bloods are high, that I don't want the correction that I just don't put it in. But, I know there are times when I do it with loads, but I can't think of any examples.

P22

I think that I have only just realised how often I actually override um my pump. Obviously, like everyone else has said there are so many factors that affect our lives that the technology we have, it can't predict it or adjust to it. One that I'm just thinking about now was, say I've done like a a long day at uni. It's been very on your feet, very practical. And then the walk home and then a dinner and then maybe work out before bed. My pump, it's it's not like a diary. I can't tell it every single thing I've done, and every step I've taken, etcetera. And so, then that's when I take it into my own hands. And yeah, it is only literally right now that I've realised that I I probably do that every single day. Whether that means maybe I need to look at my settings and adjust some things I might need to do that. But like everyone else has said, there are just certain things in life that I think we have to take control of and that the technology will never, well maybe, but will never be able to help with.

Interviewer

Yeah, I think this is, yeah, we're moving onto another area, I'd like to talk to and talk about. It's this idea that there is all this extra information that could be out there that, well it sounds like you will factor into the decisions and we'll make adjustments for. How would you feel about having to input that extra information into a system so that it's began to sort that for you? What I'm trying to get to is that kind of the link between the extra effort you'd have to put in compared to the benefit you'd get in terms of management.

P22

I think in theory it would be amazing and obviously the the control would probably be a lot better, but I think practically day-to-day there are so many like unpredictable things that happen. Like one minute you're here and then you have to rush off there and there's not always that time to input um that information. Um. But like I said, in a perfect world, that would be amazing, but for my lifestyle at the moment, anyway, I don't really have that time often to that I would be able to do that sort of thing.

P24

Yeah, I'm I think I'm the same. Like, I barely remember to turn on exercise mode when I'm about when I'm like, standing in the gym. And that's just one button. Sometimes I do, but like not every time, so I don't think I'd be able to remember something more complex every time. Um. And it would like, if it was implemented as a feature, it would have to be reversible because like ideally you want to do something like 20 minutes before you start the activity. And the other day I drove to the gym, which is only a 10 minute drive, got there and it was shut, so if it was irreversible and I thought, "OK, you're gonna be working out for two hours", that would be quite damaging because I went and got McDonald's instead.

P01

I like getting. I like that you got McDonald's instead of going to the gym that was a good move. But I think I would quite in theory I would quite like it as well because like I'm having to put the effort in anyway in like thinking “ohh do I need to change the bonus amount? Do I need to add more? Do I need to take it away?” So, if it's like the same amount or less effort for potentially better control, because in some situations I trust that technology more than me because I'm tired and busy and probably not doing things to the best of my ability at the moment. Um. But yeah, it would definitely have to be reversible and I think it would have to be something that you can do when you've got the time, but it's not going to like majorly ruin everything if you don't do it. Like it needs to be a bonus thing, but not something that the whole system relies on.

P24

Yeah, I remember, I used to have a pump. It was like the the pumps that they used to do that came in like black, silver, pink and green. I can't remember what they're called but they used to exist. And they had a um uh like a reduced basal thing where you could turn the reduced basal on for a certain amount of hours. I think Medtronic might have had as well I don't remember now, but you could.

P01

Yeah, my current Medtronic has a temporary basal option, which sounds similar to that.

P24

Yeah. Temporary basal. Yeah. Where like you could reduce it for a certain amount of hours. And I guess like sometimes I did do that if I was about to exercise, I'd know exactly how long I'm going to exercise for, and I put it on for an extra hour and then it would just tick along and turn off when I needed it to. So if it was something like that, I did actually used to do that, so maybe it would be realistic that that's something I would do now as well. Like I can imagine going on a night out and as soon as I get to the club turning it on and not and having it on until I wake up the next morning and that would help reduce lows. Or like maybe even just prepare the pump for the fact that I'm gonna have bloods all over the place or whatever.

P10

Yeah. No, I think also quite a lot of the stuff I do is quite routine based, so like whenever I exercise I always exercise the same day at a very similar time. So even if it's not kind of like a I do it 10 minutes before like even if it's a daily setting which obviously you have to go turn off as well. But I think it could be a certain amount of input with just a bit more precise, kind of level set. So, like if again if I'm going slightly higher at uni. If I was able to put any information that I want to be slightly high at this time of the day, just more precise kind of things. And I also think kind of that integration obviously like we're on the trial with the Fitbits. If that had a way of. Integrating with the pump. So, you kind of like, say, how much exercise you've been doing so, you didn't even need to put it in in the same way. Because like I always forget the exercise function, so I don't know I would with information that if there was a way to automate it out, this did make everyone's life so much easier, but still have the same, if not greater effect because it's got less human error.

P24

Yeah, I I think that like automation is really important. I remember like one of my old pumps again used to have like you could have multiple profiles and they suggested like “oh, you have a sick day profile and a summer profile and a winter profile”. And I never set any of them up because it was overwhelming. Like having to do that manually. But if your pump could automatically generate new profiles. Like even if it could take temperature readings and know that if it's a hot day based on your Fitbit because it it has temperature sensors on it, if it's a hot day, your bloods are gonna be running lower or higher, however, the heat affects you, and automatically put you in one of those profiles. Something like that, it like it could be so beneficial.

P22

I think that um as diabetics, we all have so much to think about 24/7, no break. So, I think this sort of technology would be amazing, but like other people have said, as long as it's either the same amount of effort or less. I think more effort on top of what we already have to do could just become overwhelming and then that's just a downward spiral. And then that affects your levels. And then it's just a vicious circle of terribleness. So, if it was. Or or like [P24] was saying about you, you don't even having to input anything or input minimal amount and then it using the the information it gets to then create its own sorts of profiles. That would be good, although then there would be the danger of it maybe malfunctioning or misinterpreting information, and then that's literally our lives in its hands. So, there could be a danger there. But in theory, sounds great.

Interviewer

I think as part of that as well in this extra effort you have to put in, one that one thing that I found particularly moving on to the closed system and that benefit I've seen in terms of management that I'm hitting targets that I was kind of looking for. Is that suddenly I I'm less inclined to put in that bit of extra effort because I'm kind of happy with where things are and I don't necessarily want to get things to be perfect. I just want to manage to get by. I wonder if like, has anyone else seen similar effects in the way they've gone about things that they've seen reached a point where, yeah, you kind of begin to think less about what exactly you need to do or you kind of slack on some of the other bits you're currently doing because you've hit a point you're happy with or are less inclined to do more.

P01

I've I definitely slack on things, but it's I I'm definitely not reached a point that I'm happy with, if anything, for me it's more of like the reverse where like until recently where I found out I can potentially get a closed-loop in the future. I've been asking for so long to get one and just getting absolutely like nothing back from my diabetes team that like I was sort of like a bit kind of defeatist about it. And like, just tired and fed up with dealing with it and then probably putting in less effort than I should just because I'm busy and tired and frustrated with not getting the systems that I want.

P22

Out of all the pumps that I've had and injections, I found the t:slim and Dexcom combination is the least amount of thinking I've had to do and also my control has never been better. And so sometimes I genuinely even forget about my diabetes, which is insane to say because how can you forget about it? But that's the the power of the t:slim. And so, I feel like I do sometimes slack on certain things, but I don't I haven't seen a huge negative effect from it because the technology is so good. So, um it's sort of why fix something that if it's not broken sort of thing. But if there was something that could mean even less daily overthinking and stress then I would of course try that out, but currently I I feel very content where I'm at.

P18

I am feel content now, but I have to say in relation to the the journey, like you're saying [P01], like it's been a long road. And I think that exhaustion has sort of made you not want to have to do too much with your diabetes once you've got the technology that's there. And I will hold my hand up and say I am a little bit blasé now because it is a bit more of a safety net. But to be like honestly the the way I got my closed-loop system was actually a medical emergency anyway, and a new career change, which meant I had to have it. So, if if that hadn't happened, then I would still be like in that exhausted frame of mind. So, now that it is here, you just wanna sort of rely on its technology. But I can see what you're saying in that, you can't have anything more complex or that requires you to do more because, you know, there isn't that capacity to do it. Diabetes is just too complex or too demanding as it is. And I think the the pharmaceutical companies know that and the manufacturers know that, so hopefully we'll get there.

P01

I think in relation to that as well, I feel like a sense of like like I'm owed a break from it and I feel like like I've been dealing with it for so long, like I deserve a break from it. So as soon as the technology does allow you to have that it's very hard not to like, relax a bit and and sort of try and give yourself a break, even if it ends up having negative consequences.

P10

Yeah. No, I'm similar. I kind of like I am satisfied with where I am with the pump, because I actually I've run slightly higher with the pump than what I was doing before. But I'm fine with that. Like I've just taken a lot less effort and I'm just being content with that. And I know I could probably like spend loads and loads of time trying to fine tune my pump ratios, because I'm in the quite general. I don't really use the hour by hour, I've got like 3 different settings for the whole day. And I used to like. When I first first diagnosed, I had really tight control. I had loads of effort and like had like I had the hour-by-hour ratios and they were like ever so slightly different by like a few grammes, kind of. Um. But now I've just kind of got to the point where the pump does lots of it's for me and I'm still running it like perfectly heavy levels. And that's like enough for me now rather than like putting in twice the amount of effort for like 5% better control or like just two like 2 units lower HbA1c or something like that. Like I think there's a certain extent you can do and I think the pump does lots of that for you. And then once you get past that, it just feels like so much more effort for like not that much tangible

gain in the like the short term. I know long term it probably will have a like a large effect, but just it's difficult to kind of see that.

P22

Think as well it it's not just diabetes stress that we're constantly thinking about. There's obviously like normal stress of just daily life on top of that. Um. So once again, any extra stuff, it might be like a the normal person would struggle with extra stuff, but we've already got extra stuff, so extra stuff on top of the extra stuff. Too much extra stuff.

Interviewer

Yeah, maybe going back. Well, maybe at subpoint there I think it like, it's interesting that that rate, you know there is a level that it's so personal as a condition and people have to find their own level that they're happy with and that balance isn't the same, nor should it be the same for everyone. Um. And I want to then think back about this, yeah, this extra information you could feed into it. And the the idea of being prompted for it. If there was a system that kind of asked you at different times to give maybe a little bit more information, how yeah, people would feel about that system?

P22

I think obviously if it was convenient in the day and if I if I got prompted and then I had that time to to put it in, I would have 100% do it. But then there's the worry is if if I if I didn't put that information in then would it, I don't know, would it mess up the system or would it ask. When would it next ask me? Would it be 10 minutes later? Like what if I can't do it 10 minutes later? That sort of thing. So, it I guess it would have to be very clear on, you know, if I can't put the information in now like, do ask me again but with a decent amount of time sort of thing.

P01

Yeah, I think that's a really good point, because sometimes you're just busy and you can't do anything about it. And I find that with the blood sugar alarms, like when with the Libre, when they go off like constantly like, like really, really close together and I've already put the insulin and it's like, “well, there's nothing I can do about it now”. If it's like doing a similar thing to that where it's just repeatedly going off when it's not the right moment to deal with it, then it just adds to the general frustration of diabetes management. But if it, like, goes off at a time when you do have the time to do it, I think a prompt would mean that I would make better use of it than if it was reliant on me to remember to do it.

P10

Yeah, no, I agree. I think the prompt would help. Yeah, I'll definitely forget something that I was like consciously trying to do. If it was prompt and like I had time, I would see it and do it. But I think it it would have to be a kind of system where the prompt was a bonus almost. So, if you kind of answered

it, it would be able to work better, but it would have to be like, if you didn't, that was no, by many of the end of the world, it would just revert to normal. But like I agree that if you could potentially, if it prompted me and I was just kind of like in a bus or something, or like on the train going from my side and just had it was sitting down somewhere and was just kind of waiting for something. Um. Even if it was just kind of like entering a number quickly, something like that um, I'd definitely be able to do it quite regularly. Not regularly, but whenever I was on my phone. Um. I think that would work well, but again you wouldn't be able to reliably get that information from me, at least constantly. Just the kind of way I use my phone, like I'm on it sporadically rather than regularly.

P22

I think what could be quite useful is if there was a function where you could go back and say what you were doing or um input information earlier, which it then could use to like predict or in the future. So, say if you do have a routine and and you do do the same walk or the same thing at similar times every day, even if you didn't input the information at that moment. If you have gone back and said you know, "most days I do this then", then it could learn from that and then it could maybe learn. If, if the technology could learn you more as a person then it could personalise the prompt times and I suppose that would be more useful for people who have quite regular routines. But yeah, I think if there was a way for the technology to learn you better that could be quite useful as well.

P01

Yeah, that that would be amazing if you could like link it to your calendar or something like that. I know sort of the more things like the more extra information you add into it, the more sort of like opportunities there are for it to go wrong, but I think it also comes at like the benefit of, like you say, the more it knows about you then the more it can act like a real pancreas and do its job better.

Interviewer

Yeah, I I wonder, is part of that, maybe another level. And I think that is a really interesting idea that the yeah, that the system could learn times you're more likely to respond to these prompts and then adjust them accordingly. Um. Yeah. In terms of being like maybe rewarded for these prompts, I wonder how people feel about that. And by rewarded I, the ideas I kind of could think about would maybe being told that this is the difference it might make in terms of improvements to management, just by giving that bit of extra information or giving feedback more generally about your life and how things are changing. Similar to maybe the way the Fitbit does in terms of the number of steps you do each week and how that changes. If it kind of collected that information then gave it back to you in a meaningful way. Would those kind of ideas affect your willingness to give that extra information?

P24

I mean, Dexcom used used to have that with Clarity and um I I always liked that like being able to see like oh, I did 1% better this week, that's that's nice. Um. And that did make a difference to my like, it would make me want to do better. But at the same time, it was really disheartening when I'd maybe forget to check on it for a week and then notice, like, "Oh my God, it's gone down so much like my

time and control time in range". Um. Yeah. I just think it's all really difficult because you're not gonna be able to cope with diabetes the same amount every day. Like sometimes it just makes life so much harder, if you're getting told what you're doing wrong, or even what you're doing right. It's like yeah, "great, I did that, right, shouldn't have to body just sucks". Like it can be really depressing. And some days you're going to be you're going to be ready to go and ready to make changes or just like do the right thing. And some days you just, sometimes you just don't wanna be a part of it at all.

P01

Yeah, I think that's a really good point. I would definitely feel like some days it would help me, but some days it would add to the, like guilt and anxiety and the burnout of it. I think if that was a feature, you would definitely need to be able to turn it off and on because for some people that would work really well. But also, if you're going through a phase where you're feeling exhausted with it all. Then you need to be able to turn it off if it's just going to make it worse.

Interviewer

Yeah. I I think those are really interesting points. I wonder if anybody else got kind of views on the matter or?

P10

No, I think the kind of quantitative feedback would be quite useful in a way. I think the kind of idea of the Fitbit esque kind of thing, I find that quite useful. Because if it wasn't like sent to you or anything like that, if you could kind of check it as and when. Because I I find I don't really check the Fitbit app that often, but when I do it, it's quite nice see like all I've done X amount more this week. Um. But I also feel like with the prompts, going back a bit in the conversation. I feel like it wouldn't be, it would be kind of a weird thing to do, but it wouldn't be impossible to kind of see actually get kind of semi realistic value because you entered this you were this much lower. Because obviously I know my day-to-day is quite similar. So, like if I see the difference to like an average day to what happened where did the prompt and it had this much information and then it adjusted. If it was exactly what it did because of the prompt and the impact that that had. As long as it was kind of a useful impact and it actually did something, it kind of takes it beyond just kind of this meaningless thing. Because sometimes, like the exercise function, I find I never notice it. I always forget to do it, and when I actually when I do go on a run and stuff, I always find it's too low and I just take my top off because it there's no insulin anyway because I go down quite a lot when I run. So, I never use it for that reason. But like when I am using the exercise function slightly lighter exercise, I don't notice a difference. So, if the prompt, it was kind of giving you, if the reward it was kind of giving you to see, like, "OK, it had this benefit". But it actually had kind of a it showed a quantitative slash, tangible kind of difference rather than just this kind of like "gold star, you entered something" you could actually see the difference it made, it might be quite helpful. But then I also agree, you'd want it to become positive, because sometimes you just don't have the energy, and if your thing is kind of not telling you off. But if your technology is whining at you all the time, literally just beeping at you, you just gonna get sick of it and can't remove it. So another notification telling you how bad your day's been when you know you know, it already probably isn't gonna help. So, it'll have to be that kind of like consideration.

P18

I would say the same because you only or should only access something if you feel like you're in a good place to access it, because sometimes it can be seen as sort of like constructive criticism over your own like grasp on your control. So, they should either like compile a report or like an interactive graph that you can look at your, look at in your own time or at your own leisure. Um. But one of the things I'd say is that rather than having to go through the spiel of contacting your medical teams or like your nurses or what have you, there should be like, or it could be like a little prompt that says "Ohh you did such and such this week. Well done. But maybe this slight tweak and like a a numerical tweak would improve it further". And you could either take that to your team or you could do it in place of the team. Not to, like, undermine them at all but sometimes it it can be quite laborious to laborious to go to them with something so small when it can be just a quick fix that a lot of the time I'm sort of clueless about and it's only trial and error that can improve or or make it worse. So you have to to have a bit of scientific basis to it, I think.

P22

I completely agree with that. I I find sometimes I I can. I think I can sense the pattern and then I'm like well, maybe I can change this but there's always that doubt. I don't want to be the reason that I mess up everything and then ruined my day with bad levels, etcetera, so. And like you say it it it's not always the easiest to get in contact with your team over a small little tweak. So yeah, I think I think that side of it would be really, really useful and it will be useful as well in in the sense that you you you then do what you want with that information. So, it doesn't change it it itself, you can decide whether that's a good idea or not, and you can take it to your team and talk it through. Because if if the technology just changed it by itself, then there is that danger of the fact that it can't take everything into account that's going on, that you might know more about.

P10

Yeah. No, I I agree with you completely. Like, I think one of the things always kind of baffles me is, it always so the t:slim especially, so it suspends into the. So, like when I give a certain amount of insulin. So, like I give, I take what it suggests kind of ratiowise and and if I put the slightly wrong ratio it will just accept that anyway and then it's suspending incident. So like I get that sometimes it's because of something you did, but it always baffles me that, like the technology couldn't kind of notice a pattern overtime, that every every meal, the ratio is ever so slightly off. And it just. I don't know, like in my opinion, just basic kind of pattern recognition technology you should in theory, be able to take the data. And I I agreed that I wouldn't want it to do it automatically because sometimes obviously you might input the wrong amount of carbs because you weren't sure, you were just guessing or like you were more active. If it was able to ask you like "ohh, it seems that after kind of that your breakfast slash morning meal you're going X amount of high, but you're still getting insulin. But you you're always correcting by this amount. Do you think it's a good idea to like change your ratio by X?". And now I feel like that's not a particularly unachievable technology change. But I feel like that's probably one that would. Because sitting down, fine tuning levels is the one thing I don't do. I never look back on graphs and I know I could and I know that would probably make a massive difference. But kind of having that information. Like the hard adminy bit of looking back over the

data. And kind of getting that information dug for you and and just having suggestion you being able to kind of like “OK, is this a smart idea or not?” And having that kind of, you still have the final decision but having the kind of hard boring bit done for you might be it would be really useful.

P01

Yeah, I I definitely agree with that as well. I absolutely hate looking at the reports and it's something that I avoid, like at all costs, even though it would probably help get my control a lot better. But it's so overwhelming and it's so hard to do it. It feels like I'm at school again, doing like a statistics test, but the consequence is obviously like your your actual life. So, it's really overwhelming. And if there could be technology that could like recognise these patterns and make suggestions towards it, it would be such a benefit. Even if it was just something you did to like sort of tie you over between clinics or between seeing your team, um I think that would be amazing.

Interviewer

Yeah, it feels. Like a lot of people are kind of making the point that there's this tendency with technology to almost to just include more and more things, but there is a knock on effect of that and there are consequences of it. Particularly if there are things that you can end up highlighting things that aren’t as effective and what that has an effect. Um. As kind of part of that, I almost think more broadly than in type one diabetes management, I guess the focus of this a lot so far has been on that blood glucose level and you're here. I mean historically it's HbA1c level, but now time in range is kind of this big target that you're aiming for. Do you think that that is reflective of what your aims are in terms of diabetes management. Or do you think there should be other things that should need to be taken into consideration that either you do put currently or you think should be more taken into account?

P01

I I think one thing for me is that it's never taken into account like the amount of time you spend managing your diabetes. And although obviously there's an expected level of time that you have to commit to just diabetes management, but like if you've got perfect control, but it's at the cost of like everything else in your life and you're it's affecting your work and you're up all night, like microdosing and doing corrections and stuff and it's just this huge burden on you. Versus somebody who's got kind of OKish control, but it's not taking as big a burden on on their mental health and their life. Then I think that's something that particularly like consultants should talk about more in clinics and should take into consideration because that's definitely. The burden of it is definitely as important as like your actual levels as well.

P22

Going on from that there, I feel like there are so many aspects of life which are, when you when you try and get your control around them, for example things like hormones. It's not just “ohh this is going on, so I'm gonna do this and suddenly everything's fine”, it's all about trial and error and I feel like that's not always taken into account. And then. And then if your HbA1c, your time in range is not

as great as it has been, it's I I, me personally I don't feel like there's the consideration of the amount of time that certain things take to smooth out, if that makes sense. Um. So, it's not like I'm I'm doing a terrible job at diabetes, there's there's other factors which aren't in my control, which I'm trying to control that is affecting it. And it's not just like a snap of the fingers, sorted. But that doesn't reflect in the overall HbA1c level.

P10

I also found that. So, I always, I still do use HbA1c as quite quite a big judgement. Um. I never really looked at that from graph, so I don't really see time in range that often, but my clinic always prefers time in range, to HbA1c. What I find that it's such a kind of general. My kind of thing, I didn't like about it was when I was lower spending so much. I agree completely that it's perfect levels across your life is objectively, worse health in my opinion. So, like when I was slightly younger like year 10 type of vibes, I was spending so much more time on it and my HbA1c was like crazy low, like it was like 37 or something like that. Um. And it actually just wasn't that healthy, I was having quite a few like nighttime hypors. Um. And I wasn't, like, obsessive or anything like it was my life was fine. Um. But it's such a single number, it's such a difficult reflection I think. Especially if the amount of data we have. Even like time and range is a much better reflection of it just because it's kind of, it reflects the fact that you're occasionally going to be higher, and as long as your range is kind of forgiving um you're doing the best you can and it kind of reflects on that. But HbA1c like you could be living being a really bad lifestyle, going high, low, high, low, high, low and it would reflect really in the same way. Or at least it is my understanding is that it wouldn't, as long as you were low the right amount of the time.

P01

Yeah, I definitely found that with my HbA1c before I started looking at time in range and it meant that every like clinic, my consultants were like, “ohh, you're like your control was amazing, you're a good diabetic”, which I hate that phrase. But I was just like trying to convince them that it wasn't good, because I was like 20, half the time and like 2 the other half of the time. And it meant that I felt awful all the time and it was stopping me from doing just life, but it averaged out to be a good number. So that's definitely like something that I found in the past as well.

Interviewer

Yeah. No, I think they’re kind of really interesting points and kind of how those numbers are used. And there are definitely flaws to them, particularly HbA1c, which I guess is why people are looking to shift to time in range, but that has its own problems associated with it. Moving on a little, I think another area I'd be interested to hear from you all about is. Because part of the work here I think we've spoken before is the idea of feeding information from the wearable into a closed-loop system that just based on it. But one thing as part of that, some of the ideas that range is there's a lot more information being stored about you and what you're doing and from sort of a privacy point of view, how you feel about potentially that information or being in one place and. Um. Yeah, just an increasing indication of what you're doing in your life and how your life is being run, held in a single place.

P24

I'm fine with that, is my take. I don't really have much more to add. Yeah, I just don't care.

P01

It's yeah, me neither. I don't have anything to hide. So.

P18

Hospital know enough about us anyway. And if it's going to be like literal tiny aspects about this everywhere to build this massive big picture then surely in the long run, it's in our best interest? Yeah, you sort of sign over those rights when you become a diabetic really don't you?

P10

Yeah, yeah.

P22

Sorry you go.

P10

Oh, thank you. Uh, no. I kind of agree. It's kind of like. There’s so much. As long as there's a trade-off involved. Because I feel like everyone's data gets stolen anyway like all of your Insta, stuff like that. Like I've got a Google Home mini, because I find it really like, useful and I enjoy it. And I know that it's not like good, but particularly diabetes data, I feel like the integration of the data will be so life changing like quite frankly. They like. I don't care if it's not stolen, obviously there is some like limit to the amount that you care. It's very different the kind of it being stored somewhere and you'd want there to be quite rigorous laws about kind of data protection. Which there is, especially like sensitive health data. But like, quite frankly, if I was benefiting from it enough, which I think I you would because the integration of the information would be so useful to have it all kind of in one place working together. Rather than you being the thing that holds it together, it just does it automatically. I think the difference it would make would be so worth it, I wouldn't even think about where my data is going, I'd just blindly accept it and just kind of like. Yeah, just be grateful that there was a way to get it all together rather than worrying all the time about what the big company is going to do with that.

P22

I was gonna say very, basically the same thing. You know you there's. You can find out basically anything on the internet about anyone. So and that that information that you can find out is way, to

me, in a way, in a sense, way more personal than sort of what, how many carbs I had for breakfast. And I I don't really care if someone knows that and like everyone has said, if it is helping, why not? And um obviously there would be laws and things like that. But yeah, it's very I think it's very different data and so there's there for me, there won't be that fear of someone finding out or or stealing it if it's going to improve my life, or if it's used to improve other people's lives, then I don't see a problem with that.

P01

Yeah, I I agree as well, like the benefit just outweighs any risk so massively. I don't really see that there could be that many that much sort of damaging things that could be done with somebody having that kind of data. Um. And I'm just so desperate to for some of the burden to be taken away that it would be worth it anyway.

P22

I guess something that maybe could be a problem is if the say it was with the Fitbit data um or something similar. If if there was a location setting and if maybe it if you went on a run or something and it it tracked your journey. That sort of information I would be hesitant about being shared. As in, you know that that leads to a whole other thing of you know where you live and your routine and that's a whole other thing. But in terms of just day-to-day carbs, exercise like little data like that I don't see an issue with that.

P01

Yeah, that's a good point actually. I haven't thought of it like that. I think with that kind of data and or even with it sort of we talked about it before like knowing you and knowing your routine, I would find it worse if, like an individual person got hold of that rather than like a big company because I feel like with a big company they would maybe use it against you in the sense that they might have more targeted adverts and stuff like that, which I I don't mind that much about. Obviously if it was something that was actually gonna impact my, like personal safety, then that would be a whole other level. But I would assume that there would be such strict rules about who could access this and that it would be like anonymised and wherever it was stored anyway that that would all be accounted for.

P10

Yeah, no, I kind of agree. No you go.

P22

I was just really quick. I was gonna say I think even no matter what was said, you know, like your your information about location etcetera, is not being shared, no one can see it, anonymous, blah, blah blah. I think that I would still be a bit worried about. Yeah, you're saying no one can access it, but,

you know, people break law, people use information in in bad ways. And no matter what you tell me, there's going to be a part of me that worries about you know where the information actually is.

P01

Although I I imagine that that people can probably already get from your phone anyway. So, in some ways that's more of a like general problem that rather than something that's directly related to this, but I guess it all it all like links to the same issue.

P10

Yeah. No, I kind of agree is in like I I think your phone does so much stuff. I've kind of like become immune to the worry. And also just the volume of data. I know that they could process it quite quickly and stuff like that, but like I can barely make sense of it, so good luck making sense of like millions of diabetics worth of data. Which I know they could just in my mind it's just like there is so much stuff to sit through and but I do really kind of like location. And the other thing that it's slightly different diabetes stuff, but the Fitbit. I know my Fitbit has a microphone on which I find kind of strange. I don't like the fact that like it has Amazon Assistant, I think that's kind of it creeped me out, so I did turn the microphone off quite quickly and yeah. But I think, yeah, location kind of microphone I think feel very different to me than kind of my glucose readings and stuff like that like. The most you can tell from glucose readings is how well my diabetes is controlled. Um. Which is personal data, but like I don't think you can weaponise that particularly well, unless it went to like insurance companies. Obviously, a bit like that, that here in the UK actually don't have that issue. And then stuff like like when I eat and stuff again, it's not particularly like weaponisable data. I think when it gets to the point where they can tell loads about your behaviour, it could be concerning that I would. Yeah, I think general diabetes data, I don't see much of a problem with kind of like my own personal privacy.

Interviewer

Yeah, that seems to. Uh. That's like a range in yeah, views on that point of view and for different parts of it. I wonder currently how control how in control you will feel with that data, in terms of that data sorry.

P10

I know I feel kind of like it's decentralised. I don't particularly feel particularly in control, especially like even this like I trust you. Uh, completely but I'm just sending you all of my data. It's kind of like I read through it and all of that bit, but I was it never kind of crossed my mind to worry about that. So, it's the same with kind of like, obviously healthcare needs are different, like I know that Dexcom has all my data and it's sort of Clarity, etcetera um and I'm fine with that. I don't feel particularly in control with it just because I know I'm sharing with certain people and exporting it out as well. So, I don't get any control, but I also don't mind as well.

P22

I think that with the amount of information we unknowingly knowingly share, like from our phones and everything. Like with this study, I I just, it hasn't even really crossed my mind to worry or even think about it. And it's only you asking that question now that I've actually started thinking about it. So yeah, I I think I would say similar I I don't really know where I stand with that and I think it is to do with the way like phones and our information in other aspects of life is shared and stored.

P18

Yeah, I don't think I've thought that much about it either. And I've always kind of had the attitude of like, I've got nothing to hide, so I don't really mind who can see any of my data, whether that's diabetes data or whether it's like just what I'm doing on my phone or my laptop or whatever. Um. Even if it's even if like my Google Home is like listening to me, I feel like nothing bad can come with that but maybe I'm just being really naive about it and I thought about it more from an angle of like whatever that somebody would find something that would get me into trouble rather than from the angle of like personal safety, which is definitely more of a concern. Because like I know what I've been doing and saying and searching and I know it's all fine but with like going back to the location stuff that is the only thing that would make me slightly like nervous, I guess.

P24

I think. Generally, I agree with what everyone else is saying and in my day-to-day life, I'm probably like I care even less like. But like I I've had. I think it fluctuates depending on where your mental health is and your mental health history, because I've had psychotic delusions and periods of psychosis in the past where my data and like the delusions meant that I wasn't feeling that trust. In fact, it was quite the opposite, I felt fear and I worry about the implications that that would have if that data was actually something that was supporting my life. Would my mentally ill brain cause me to tamper with that and have significant influences on my life? I don't know. I just think it's something that like would have to be considered like in in a case-by-case basis if this if this technology was to be rolled out.

Interviewer

Yeah, I think. With us, with so many of these things, it is personal and different people, different things. Yeah, I guess I'd maybe also as part of that, I guess they're. What we've been talking about is quite broad access to all this data in one place as well. But there are avenues that people share it currently, whether that's just like blood glucose readings with friends and family or bits with the hospital. Yeah, and I just wondered for people, how that kind of like willingness to share changes for different people changes for the different people they're willing to share it with? And also if it different occasions or their occasions you're more willing or less willing to share bits of that diabetes data?

P22

Ohh, I've got my boyfriend and my best friend on the Dexcom immediate follow app. Um. And for me I set that up for like day-to-day life and the the worry for me is night times when if if I don't wake up from a low or something like that, and if my boyfriend's there, he can help me out. Or nights out is another reason I have the them on the follow up. However, I don't actually have my parents on the immediate follow on, I have them on the the Clarity follow. So they get, they get it 3 hours later and I mean I guess it differs from person to person and depending on your relationship with your parents. But for me, I want a bit of control and independence, myself and especially being at uni, I just feel if my parents can see exactly what's going on to the minute, I lose a sense of control. And then I feel like a child again where they're basically in control of my diabetes. So for me that's why I don't have them on the the follow app. But yeah.

P01

Yeah , I I have, because I've got the Libre, it doesn't seem to have quite the same system of like letting people follow you as the Dexcom does. Which is really frustrating because I like the look of the the Dexcom system and the way that my brother uses it where it like can send somebody a location. If you've got a critical low. But the FreeStyle one, it lets people look at it but only like after you've scanned it or something like that. It it, the graph never seems to be quite up to date with what I'm doing. So, I've not really bothered to add multiple people on there just because it doesn't seem to be that useful. If I had the Dexcom one I think I would definitely do what [P22] was saying and add, like my boyfriend and my friends. Because there are like situations like night times or nighttime hypers and stuff where it's really beneficial to have somebody be able to access it from their own phone. At the moment, I only have my mum on my Libre one, which is generally fine. But like you were kind of touching on, that's been like the odd occasion where, like I felt very sort of yeah, like I'm a child and like she's checking up on me again. Which isn't necessarily bad, it's just. It's not like I'm trying to hide and I think either it's just sometimes you want to feel a bit more independent. And also, I feel like it's adding to her anxiety more than anything. Um. And I like, I don't want to be more of a burden.

P10

You know, I think I'm quite since I've got my I've got Dexcom and and like, have my mum following and that's it. And I always feel like sometimes it makes it so much worse. So, like I don't use my phone that often for Dexcom. So, like after a while it does kind of go off. Like if I'm using a week or something, the application we just turn off it itself off like I don't have it on. Is it like I've got the settings, so it should in theory, still in the background but if I'm not opening up for a week, it will not last that long and the notification was shut off and it would disconnect. And then like sometimes I'll get really like weird text from my mom saying, like, "are you alive?" She's just, like, being dramatic. But it I find though I'm pleased I share that information, sometimes I do find it frustrating even that kind of like worry it can cause. Some I'm fine giving out that information, but I almost want it to be like a one way thing. And I get that kind of a person person discussion kind of vibe, but I'm fine with with giving out information as a just in case, but I almost don't want it to be like acknowledged in any other way. It's kind of an odd relationship with that, but that's just the way I feel about it.

P01

Yeah, it definitely has a weird adds a weird tone to relationships because it makes you feel what makes me feel like a patient. And because my parents like were responsible for my diabetes when I was a child, I feel like I don't want to revert and go back to that. Um. And like, yeah, I I want my parents to be able to just trust that I'm managing it, OK. And when they've got the option to like to check on it. Sometimes it's fine, but sometimes it does feel like a bit of a like like there's a lack of trust there, if if they're choosing to check on it outside of emergencies.

P10

Yeah no, that's quite interesting because I was diagnosed quite late, so I've always been kind of like in sole control to the point where, like my mum gets it, she was, you knows, quite a lot about it, but she doesn't understand it anywhere near to the level you'd have to be to control it for someone, which is kind of an odd vibe. Because I sometimes think is the sharing the data makes it worse because it's kind of the numbers, not meaningless by any stretch of the imagination, she understands plenty. She’s obviously done research and stuff like that. But like, because like from year 10. So I was managing it completely by myself, like I've never had her injecting or anything like that. Like it's always been me monitoring my levels, so I just kind of talk to her about every now and again. I feel like it almost makes it more confusing because I feel like they're sharing the data of the person you know, even instead of big company, you get more judged more on how are you doing as a person. Especially when they send text messages saying like, “well, you do this, this and this”. So, it feels more personal when you send data. I think to individual.

P01

Yeah, I think I, I actually I felt like that even though my parents managed it when I was younger, because the technology has changed so much since the time where they were managing. Like when my younger brother got diagnosed, it was the mixed insulin where you had to, like, do it at a really set time and it meant you had to eat the same amount of carbs that are really set time. And obviously they learn to manage diabetes with much worse insulin and much worse technology on a 2 year old that is much harder to control on. And then applying that to like an adult with much better technology and more of a like grasp on it. Sometimes they, like, don't quite understand it as much as they might think they do, and I always think like a little bit of knowledge can be more dangerous than no knowledge.

P18

I'm just like thinking through all of the processes I've had to like link up and create accounts for and because I've had I think 3 different kinds of sensors now, I've actually got to the point where there's nobody attached to my share or like Carelink it's called for Medtronic. Nobody is attached because it's just too much of a faff. But I think if I were to look at it, it probably wouldn't be parents either, because like as you're saying, it's that relation to being a child again or not in sole control of your diabetes, which can feel quite belittling. So, in the best way possible, of course but it's just it's good to have that independence. That being said, I can totally see why it's still relevant to have them for safety. If you're university or you're doing things that could be could make you particularly vulnerable. But as I said, like it's just been too much of a faff to link it all up this time to the third or fourth sensors I've had.

Interviewer

Yeah, that's really interesting to hear across all those those cases. Um. I guess uh, one final and this just a quick thing before we finish off is. Just kind of was previously if there is anything interesting in the last month of using the smartwatch as part of uh, the type one diabetes management or link to this type one diabetes management. Um. And just how that's kind of changed in the last month and if there is anything interesting, you've seen.

P01

One one thing I can think of for me is I chose the Fitbit that doesn't allow you to have like blood sugar notifications come up on the screen. Um. And recently I like managed to really badly like completely break my phone the night before I went on holiday and it was really really stressful. And that did make me wish that I had some kind of like back up to still be able to check, like, see what the Libre value was with a watch or something. That's not really something I've thought of before, but it definitely made me think of like there's even more advantages to connecting wearables to the diabetes management than I'd sort of considered before.

P10

Yeah no, I find it quite interesting cause I've just done uni and move to uni and it's quite a walkable city. So, I'm walking like 25,000 steps a day and it doesn't feel like. Like, it doesn't feel like I'm doing that much exercise like having the Fitbit there, kind of it makes me aware just how much more exercise I'm doing. Because I really am doing insane amounts more than I was at home. Um. And that paired with like being ill, it's just it's a weird combination of my levels going low then high, and then I keep changing around around ratios. I think the Fitbit is making a bit more aware of that a bit more forgiving, because it's letting me know just how much more I'm doing. And it's kind of like “OK, I don't need to be stressed about this, this is just my malfunctioning or this is my levels going nowhere, it's just me doing more exercise. It will just take a few more times to get the ratios working.” So it's been really helpful in that way actually.

P22

I've recently had some issues with my heart and which have in turn affected my diabetes control and yeah, made it a bit more unpredictable. However, I've found the the heart rate thing on the Fitbit has really helped and with the Fitbit, it's affected by my activity levels as well. And so being able to compare how I'm feeling with the exercise and the steps with my heart rate. I have actually in the last month found that quite useful, and that's something that I never thought that I would have to. Have a need for so that's been good.

Interviewer

No, that's that's brilliant. And those are some really interesting points.